Lessons Learned

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Right now is a good stopping point for this project. This document reviews the project, the lessons I learned, and my plans for the future.

What was built

A wearable insole that tells a rehabilitation patient what percentage of their body weight they are putting on an injured foot, in real time.

Two force-sensing resistors sit under the heel and the ball of the foot, wired as voltage dividers into a Seeed XIAO nRF52840. Even though the device reports pounds, it is less accurate. Instead, the ratio is accurate: the patient lifts the foot with the insole in their shoe to set a zero point, stands on it to set an upper bound, and every reading after that shows where the current load falls between those two anchors. It runs on a LiPo, transmits over Bluetooth, and buzzes when the patient crosses their prescribed limit. All the electronics fit into an ankle strap.

The project started because my mom broke both legs and was told to stay at a percentage of her body weight, with no way to know what that felt like besides the inaccurate “bathroom scale” method doctors told her.

Strengths

Every reading lands high, which is the direction we want. It would be bad to tell a patient they are only at 30% when they are really at 50%. Now, after fixing the startup to record the median of a steady hold, not the mean of a wiggly one, it has a predicted average error of 3.88 percentage points.

The at-home calibration held up. When I recalibrated the heel sensor directly against the force plate, the new curve came out nearly identical to the one I had fitted months earlier on my bathroom scale. That matters for the future: I can iterate at home in about twenty minutes without lab access.

It is wireless, it beeps, and it is sleek. Bluetooth, battery, buzzer, and microcontroller all fit into an anklet.

Weaknesses / Plans for the Future

The calibration asks the patient to do the one thing they cannot do. The upper bound is set by loading the foot to full body weight. A partial weight-bearing patient is, by definition, someone who cannot do that. Unfortunately, I didn't realize this until the end. This is the device's biggest flaw, but I've started to work on it. This is just a user-interface problem, not a measurement problem - the device measures correctly; the startup procedure is unusable. My plan to fix this is to offer 2 different options to patients inside the app. The first option is the best: calibrate outside the shoe with the good foot standing on it, then add the lacing preload back in to set the upper bound. Calibrate at the prescribed limit instead of 100% (if the patient has both legs broken), pressing the foot against a bathroom scale. I already started to test this based on the lab data: worse across the full range, but better where a patient actually is - mean error 1.76 points under 30%, and it never banded low.

The ball sensor is inaccurate: The ball sensor is clearly the worse of the two from the heel sensor with an R^2 of .64. To fix this, I will order more FSRs and replace it to make it more like the heel sensor

Other plans:

- Build a proper iPhone app instead of reading the device through a generic Bluetooth terminal. Also build an interactive display
- Better insole material than leftover conductive foam, to make it more durable
- Reach out to the FSR supplier to establish a relationship
- Find a faster calibration path. Even though the at-home method works to calibrate the sensors, calibrating every sensor individually does not scale to mass production
- Test on an actual patient

What I learned

Failing is necessary. I never would have found the calibration bug if the stand-on-one-foot trials had not contradicted themselves. Everybody says there are ups and downs in engineering, and I did not realize how much that hits home until I experienced it myself.

The path you think you're on isn't where you end up. At the start I was completely set on conductive foam. I loved the setup, copper tape on both sides, making my own sensor from scratch. It drifted, and I ended up on commercial FSRs. That's how it sometimes goes.

Sometimes, when you are so focused on one thing, you neglect the simple solution staring you in the face. This was especially prominent when I couldn't get the buzzer to work on the perfboard. I spent hours testing different solutions when, in the end, all I had to do was resolder the microcontroller in. Or during the project, when I was so focused on the current task that I neglected that the setup procedure asking the patient to stand completely on their leg is something they could never do.

Going into this project, I was essentially at zero in my understanding of biomechanics, electronics, and instrumentation. I have learned a lot, such as the ground reaction force on your foot, the double hump walking pattern, how doctors prescribe PWB, what gait analysis is, how to solder, how to wire a breadboard, how to build a voltage divider, how to get Bluetooth working on a XIAO nRF52840, how to read force-plate files and time-align two data streams recorded at different rates, how to build non-linear calibration curves, and many more things.